AD33/AI33

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RAMAIAH Institute of Technology

(Autonomous Institute, Affiliated to VTU)
(Approved by AICTE, New Delhi & Govt. of Karnataka)
Accredited by NBA & NAAC with 'A+' Grade

SEMESTER END EXAMINATIONS – APRIL 2023

Program	:	**B.E. – Artificial Intelligence and Data Science / Artificial Intelligence and Machine Learning**	**Semester**	:	**III**
Course Name	:	**Data Structures**	**Max. Marks**	:	**100**
Course Code	:	**AD33 / AI33**	**Duration**	:	**3 Hrs**

Instructions to the Candidates:

- Answer one full question from each unit.

UNIT - I

1.	a)	Define a structure called *Student* with the member's *student_name, RollNo, admission_date* and *Dept,* where *admission_date* is a nested structure. Write a C program which uses functions to read the data of 10 students and display data department wise.	CO1	(08)
	b)	Explain *strncpy()* and *strtok()* functions with suitable syntax and examples.	CO1	(06)
	c)	Illustrate the dynamic allocation for one dimensional array with a suitable example.	CO1	(06)
2.	a)	Write a C function to extract a portion of a character string and print the extracted string. Assume that *m* characters are extracted starting with the n^{th} character.	CO1	(08)
	b)	Write an ADT for a complex number and include necessary operations.	CO1	(06)
	c)	Write a C function that locates the last occurrence of an entered character in a string without using any built-in string functions.	CO1	(06)

UNIT - II

3.	a)	Write postfix form of the following expressions using stack in tabular form: i) (d-c)/e+f/(g-k+l-i) ii) (b*b)-(4*k-l+(r-w)-h*o)	CO2	(08)
	b)	Write C functions to perform add and delete operations on a circular queue using array implementation.	CO2	(08)
	c)	Differentiate between Stack and Queue.	CO2	(04)
4.	a)	Explain the role of stack as a system stack with a neat diagram.	CO2	(06)
	b)	Write a C program to evaluate the following postfix expression **2 3 1 * + 9 –** Tabulate the input, stack contents and the output.	CO2	(08)
	c)	Illustrate operations on multiple stacks with a neat diagram.	CO2	(06)

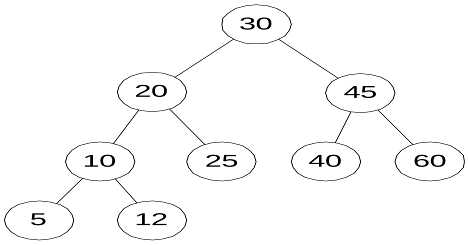
UNIT - III

5.	a)	Write a C program to add two polynomials using linked list representations.	CO3	(10)
	b)	Write a C function to perform add and delete operations in linked queue.	CO3	(10)

6.	a)	Differentiate between Singly linked list and doubly linked list implementations.	CO3	(06)
	b)	Write a C program to implement a FIFO queue using linked list.	CO3	(06)
	c)	Write a C function to invert a singly linked list.	CO4	(08)

UNIT- IV

7.	a)	Define with an example each of the following: i) Binary tree. ii) Full binary tree. iii) Complete binary tree.	CO4	(06)
	b)	Write the Pre-Order, In-Order and Post-Order traversals of the following binary tree.	CO4	(06)



Binary Search Tree T_4

	c)	Describe satisfiability problem. Justify how it can be solved using binary tree with a suitable example.	CO4	(08)
8.	a)	Write a recursive C function for in order traversal of a binary tree.	CO4	(06)
	b)	Suppose that we have the following key values: 7, 16, 49, 82, 5, 31, 6, 2, and 44. Write out the MIN heap after each value is inserted into the heap.	CO4	(06)
	c)	Write a C function to perform Right insertion in a threaded Binary Tree.	CO4	(08)

UNIT - V

9.	a)	Write a C function to create a graph using an adjacency matrix representation.	CO5	(06)
	b)	Explain the relation between degree, vertices and edges in a directed graph with a suitable example.	CO5	(06)
	c)	Write an ADT for a Graph and include necessary operations.	CO5	(08)
10.	a)	Write necessary C functions for breadth first search of a graph represented as an adjacency list.	CO5	(06)
	b)	Suppose that we have the following key values: 14, 24, 9, 5, 20, 35, 44, 7, 11, 8. Write out the AVL tree after each value is inserted into the tree.	CO5	(08)
	c)	Explain RL rotation of AVL with a neat diagram.	CO5	(06)

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